INCH-POUND
MIL-DTL-24146A
3 February 2023
SUPERSEDING
MIL-H-24146
2 June 1965

DETAIL SPECIFICATION

HEAD, SPRINKLER, FIRE PROTECTION, FOAM-WATER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers foam-water fire protection sprinkler heads for upright or pendent installation in overhead sprinkler systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-PRF-24385	-	Fire Extinguishing Agent, Aqueous Film-Forming Foam (AFFF) Liquid Concentrate, for Fresh and Sea Water

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM DS56L/SAE HS-1086	-	Metals and Alloys in the Unified Numbering System (UNS)
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. Sprinkler head bodies shall be made of bronze alloy UNS C23000, C83600, or C92200. Deflector plates and other fabricated parts shall be naval brass alloy UNS C46200, C46400, C48200, or C48500. Supporting arms shall be of corresponding or other materials resistant to corrosion and high temperature.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Construction. The foam-water sprinkler shall have a single open orifice, an agitation device, and a deflector plate, all firmly supported. The deflector plate shall be supported by one cross yoke of 5/32-inch maximum thickness designed to produce a homogenous umbrella-shaped spray that will not impinge on the overhead. The body shall be designed to introduce and mix air into foam-liquid solution (see 6.3.4) in order to produce air-aspirated foam (see 6.3.3). Unless otherwise specified (see 6.2), the sprinkler head shall incorporate a ½-inch National Pipe Thread (NPT) male threaded inlet. Flat surfaces shall be provided for a wrench to be used during installation. The maximum overall length of the sprinkler shall be 6½ inches; the maximum width of diameter shall be 2¼ inches. The orifice shall be greater than ¼-inch diameter.

3.5 Marking. Each sprinkler head shall be clearly marked with the manufacturer's name or trademark, the manufacturer's part number, and the number of this specification.

3.6 Workmanship.

3.6.1 Sprinkler heads. All sprinkler heads shall be sound, smoothly cored, true to form, uniform in texture, and free from cold shunts, porosity, or any other defects that may affect their serviceability. They shall be thoroughly cleaned, both inside and outside, and all fins and roughness shall be removed.

3.6.2 Castings. Castings shall not be repaired, plugged, or welded unless approved by the inspector. The repaired, plugged, or welded defects shall be small (less than 10 percent of a size which would be cause for rejection of the heads) and shall not adversely affect the strength, use, or machinability of the castings.

3.7 Flow and foam characteristics. The sprinkler shall discharge water or a foam-liquid solution of freshwater and aqueous film-forming foam (AFFF) conforming to MIL-PRF-24385. The foam-liquid solution shall be either 3 percent of type 3 AFFF or 6 percent of type 6 AFFF, as defined in MIL-PRF-24385. For existing approved nozzle designs conforming to MIL-H-24146 dated 2 June 1965, retesting with AFFF in accordance with MIL-PRF-24385 is not required.

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3.7.1 Rate of discharge. The sprinkler shall discharge water or a foam-liquid solution in a dispersed pattern at a flow rate of 16 to 19 gallons per minute (gal/min) at 35 pounds per square inch of gage (psig) and 20 to 22 gal/min at 50 psig.

3.7.2 Average foam distribution. Average foam distribution shall be not less than 0.4 inch high in each test container in each of 6 feet of a 7-foot radius following the test procedure of 4.5.2 at 30 psig test pressure and repeated at 60 psig test pressure.

3.7.3 Average water distribution. While following the test procedure of 4.5.3, average water distribution in each test container shall be not less than 0.06 gal/min/sq ft in each of 7 feet of an 8-foot radius at 30 psig, and not less than 0.07 gal/min/sq ft in each of 7 feet of an 8-foot radius at 50 psig.

3.7.4 Foam expansion and drainage. With foam-liquid solution at 68 °F and 35 psig and following the test procedure of 4.5.4, the expansion (see 6.3.2) of the foam shall be at least 5.5 and the 25-percent drainage time of the foam shall be at least 55 seconds.

3.8 Fire. With foam generated from a premixed solution of either 3 percent of type 3 or 6 percent of type 6 AFFF conforming to MIL-PRF-24385 and freshwater, the sprinkler heads shall be capable of passing the fire tests as described in 4.6. For existing approved nozzle designs conforming to MIL-H-24146 dated 2 June 1965, retesting with AFFF in accordance with MIL-PRF-24385 is not required.

3.9 Heat. The sprinkler heads shall be capable of passing the heat tests as described in 4.7.

3.10 Mechanical shock. The sprinkler heads shall be capable of withstanding grade A, class I shock requirements in accordance with MIL-DTL-901.

3.11 Vibration. The sprinkler heads shall be capable of passing the type I environmental vibration test in accordance with MIL-STD-167-1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on sprinkler heads when a first article sample is required (see 3.1). This inspection shall include the requirements specified in table I. The first article sample size shall be four. Visual examination and dimensional checks shall be performed before the performance tests, and the performance tests shall be conducted in the order listed in section 4.

TABLE I. Requirements for first article and conformance inspection.

Requirement or Attribute	Requirement Paragraph	Test Paragraph	First Article Inspection	Conformance Inspection
Material - body, deflector plates, support arms	3.2	4.4	Yes	Yes
Recycled and recoverable	3.3	4.4	Yes	No
Construction	3.4	4.4	Yes (including firm support for agitation device and deflector plate)	Yes (visual and dimensional exams only)
Marking	3.5	4.4	Yes	Yes
Workmanship	3.6	4.4	Yes	Yes
Flow and foam characteristics	3.7	4.5	Yes	No

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TABLE I. Requirements for first article and conformance inspection – Continued.

Requirement or Attribute	Requirement Paragraph	Test Paragraph	First Article Inspection	Conformance Inspection
Fire	3.8	4.6	Yes	No
Heat	3.9	4.7	Yes	No
Shock	3.10	4.8	Yes	No
Vibration	3.11	4.9	Yes	No

4.3 Conformance inspection. Conformance inspection shall include the requirements specified in table I. Conformance inspections shall be performed on a sample from each lot. All sprinkler heads offered for delivery at one time shall be considered a lot. Sample size and sample selection shall be in accordance with MIL-STD-1916, using Verification Level V in the following two tables: Code Letters (CL) for Entry into the Sampling Tables and Attributes Sampling Plans.

4.4 Visual and dimensional examination. Each of the sample sprinkler heads selected in accordance with 4.3 shall be visually and dimensionally examined to verify compliance with this specification. Any sprinkler head in the sample containing a visual or dimensional defect shall be cause for rejection of the lot represented by the sample. Examination shall be conducted as specified in table II.

TABLE II. Visual and dimensional inspection.

Paragraph	Attribute	Verification Method	Acceptance Value
3.2	Material of construction	Certificate (see 6.2)	Manufacturer certifies compliance
3.4	Single open orifice, venturi body, deflector plate, cross yoke, agitation device	Visual inspection	Required components are present
3.4	Cross yoke maximum thickness 5/32 inch	Dimensional inspection	Maximum thickness is 5/32 inch
3.4	NPT male pipe threads	Dimensional inspection	Thread form is NPT; unless otherwise specified (see 6.2), size is ½ inch
3.4	Wrench flats	Visual inspection	Wrench flats are present
3.4	Dimensions: overall length, overall width, orifice	Dimensional inspection	Maximum length is 6½ inches; maximum width is 2¼ inches; orifice is greater than ¼ inch
3.5	Marking	Visual inspection	The required markings are present
3.6	Workmanship	Visual inspection	The required attributes are satisfied

4.5 Flow and foam characteristics. The following tests shall be performed for each deflector plate type (pendent and upright).

4.5.1 Rate of discharge. The procedure for determining the rate of discharge shall be determined by the manufacturer. The rate of discharge shall be as specified in 3.7.1.

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4.5.2 Average foam distribution. Foam-liquid solution (see 3.7) shall be discharged for 1 minute through the head rotating at ¾ to 1 revolution per minute. The rotating head shall be located 7½ feet directly above the top of a single row of fourteen 12- by 12-inch open-top collecting containers. The test shall be performed at both 30 psig and 60 psig and the height of foam in each container shall be recorded for each pressure. As an alternative, the test fixture may be constructed with a fixed sprinkler head and a rotating set of collection containers. The average depth of foam shall be as specified in 3.7.2.

4.5.3 Average water distribution. Water shall be discharged through the head rotating at ¾ to 1 revolution per minute. The rotating head shall be located 7½ feet directly above the top of a single row of sixteen 12- by 12-inch open-top collecting containers. The test shall be performed at both 30 psig and 50 psig and the application density in each container shall be recorded for each pressure. As an alternative, the test fixture may be constructed with a fixed sprinkler head and a rotating set of collection containers. The average water distribution density shall be as specified in 3.7.3.

4.5.4 Foam expansion and drainage. The foam-liquid solution and the foam quality shall be as specified in 3.7 and 3.7.4. The expansion and drainage shall be determined using the drainage equipment shown on figure 1 and two sprinkler heads placed 8 feet apart and in their intended installed position. The foam collector shall be positioned so that the top of the receiver is midway and 8 feet below the bottom of the heads. Two pans shall be used to collect the draining liquid. The standard sieve receiving pan shall be used to collect the liquid draining during the foam application and prior to the weighing operation; the drainage pan, a metal pan at least 9 inches in diameter and about 1 inch high, shall be used to collect the liquid draining during the weighing operation. Prior to test:

a. The total weight (W_1) of the sieve plus the drainage pan shall be determined,

b. The weight of the receiving pan (W_3) shall be determined, and

c. The drainage pan shall be positioned on the laboratory balance and the balance weights then set to indicate the weight of the drainage pan.

Before the foam application, the sieve shall be wetted with water and then shaken to remove excess water. The sieve shall be nested in the receiving pan and both placed below the funnel. Foam generation shall be started and after steady flow has been obtained, the sliding cover shall be removed. When the sieve has been filled with foam, flow to the sieve shall be cut off by means of moving the funnel so that the foam shield covers the sieve. Timing shall be started at this point. The sieve and the receiving pan shall then be separated. The receiving pan shall be placed aside for later weighing, and the sieve shall be supported on a slope of 1 to 14 above the drainage pan which was previously positioned on the balance. The sliding weight of the balance shall be advanced in 10-gram increments and the times noted at which the beam balances. The weighing shall continue until the total weight of the drained liquid, in the drainage and receiving pans, is at least 25 percent of the total weight of the foam sample. The total weight of the foam sample (W_F) shall be calculated as follows:

$$W_F = (W_2 - W_1) + (W_4 - W_3)$$

Where W_1 = weight of sieve plus drainage pans, grams.

W_2 = weight of sieve with foam, plus drainage pan with drained liquid, grams.

W_3 = weight of receiving pan, grams.

W_4 = weight of receiving pan with drained liquid, grams.

The total weight of the drained liquid is equal to the sums of the weights of the liquid in the receiving pan and the drainage pan. The 25-percent drainage time (see 6.3.1) is the time required for drainage of a weight of liquid equal to 25 percent of the total weight of the foam sample ($W_F/4$). The 25-percent drainage time shall be calculated by interpolation of the weight-time data obtained. The expansion shall be calculated as follows:

$$\text{Expansion} = \frac{V}{W_F}$$

Where V = volume of foam sample collected in sieve = 1650 milliliters.

The expansion of the foam and the 25-percent drainage time shall be as specified in 3.7.4.

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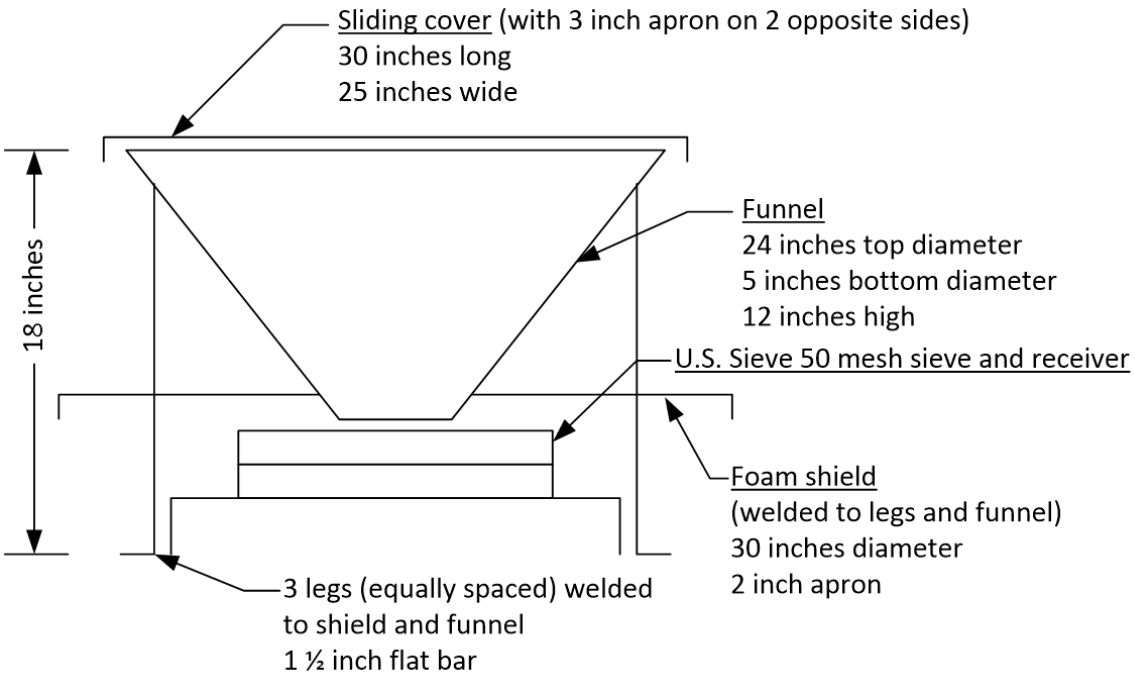


FIGURE 1. Foam collector.

4.6 Fire tests. The following tests shall be performed for each deflector plate type (pendent and upright).

4.6.1 Extinguishing. The tank used for the fire tests shall be of steel construction measuring 10-feet square by 3-feet deep. Four heads shall be positioned in their intended installed position (see 6.2) in an overhead piping system so that the discharge from the head shall be 8 feet above the fuel surface. The heads shall be 8 feet apart, each head above and adjacent to a tank corner. One hundred gallons of commercial grade gasoline shall be floated on a sufficient quantity of water so that the fuel surface is 2 feet below the top edge of the tank. The fire shall be permitted to burn freely for 30 seconds before foam application. Foam-liquid solution in accordance with 3.8 shall be applied for 90 seconds. The periods after start of foam application shall be recorded as required for the foam to spread over the tank, for the fire to be extinguished except for licks of flame at the edges of the foam blanket, and for complete extinguishment, as “coverage”, “control”, and “extinguishment”, respectively. Foam from the sprinkler (at 35 psig) shall have a time of coverage of not more than 20 seconds, control in not more than 50 seconds, and shall completely extinguish the fire in not more than 75 seconds. The foam blanket shall average not less than 2½ inches in thickness after the 90-second period of application. The fire test shall be conducted when the wind velocity is below 5 miles per hour. After completion of the tests described in 4.6.2 and 4.6.3, the pressure shall be increased to 100 psig and the test repeated. Extinguishment time shall not exceed 75 seconds.

4.6.2 Sealability. A lighted torch shall be passed over the 90-second foam application of 4.6.1 to include the entire foam blanket, including corners, approximately 1 inch above the foam blanket without touching or penetrating the foam. The torch test will be conducted immediately after the end of foam discharge and again after 10 minutes. Each torch test shall be conducted for a period not less than 1 minute. The foam blanket shall prevent reignition of the fuel.

4.6.3 Burnback test. Immediately after completion of the sealability test, an opening, 6 inches square, shall be made in the approximate center of the foam blanket and the exposed fuel shall be reignited with a torch. The reignited fire shall be permitted to burn for 5 minutes. After the 5-minute burning period, the burnback area shall be determined. The foam blanket shall prevent spread of the fire beyond an area approximately 20 inches square.

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4.7 Heat tests. The following tests shall be performed for each deflector plate type (pendent and upright).

4.7.1 Quench test. Two heads shall be placed in a furnace maintained at 900 °F. When the heads reach this temperature, one head shall be immediately quenched in water at 70 °F and the other head shall be cooled in air to 70 °F. The head shall not be deformed when heated to 900 °F and quenched in water or air.

4.7.2 Destructive test. The head shall be subjected to increasing temperatures in 100 °F increments and held at each temperature for 15 minutes until failure occurs. At each temperature, the head shall be struck with a ⅛-inch diameter steel rod to determine its structural strength. The head shall not fail structurally when heated to 1500 °F.

4.8 Mechanical shock test. The following test shall be performed for each deflector plate type (pendent and upright). Two sprinkler heads, one as received and one water-quenched after heating to 900 °F, shall be subjected to the lightweight shock test in MIL-DTL-901.

4.9 Vibration test. Each type of sprinkler (pendent and upright) shall be qualified for type I environmental vibration in accordance with MIL-STD-167-1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This sprinkler head is intended for use aboard ships in overhead sprinkler systems dispensing a solution of AFFF and seawater, mixed at a nominal 3-percent or 6-percent ratio. The sprinkler heads are intended to be installed in copper nickel tubing in accordance with MIL-T-16420. The military features of this sprinkler head are its resistance to shock impulse and to shipboard vibrations, as well as long life in a marine environment and when exposed to seawater. The foam generating and fire extinguishing capabilities of this sprinkler head are validated for AFFF only. Military components intending to utilize this sprinkler head with foam concentrates other than AFFF are advised to conduct performance testing to validate satisfactory foam quality and fire extinguishing performance. Foam-water sprinkler heads procured to this specification are form, fit, and function replacements for prior revisions. Existing stocks manufactured to MIL-H-24146 dated 2 June 1965 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Installed position, whether upright or pendent.
c. When first article inspection is required (see 3.1).
d. Inlet connection if other than ½ inch NPT (see 3.4).
e. Requirement for certificate of material (see 4.4).
f. Packaging requirements (see 5.1).
g. Whether drawings are required.

6.3 Definition of terms. The terms utilized in the requirements for design and performance of the sprinkler heads are defined herein. All pressures referred to are the gage pressures at the inlet to the sprinkler.

6.3.1 Drainage time. The time required for 25 percent of the liquid contained in the foam to drain out.

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6.3.2 Expansion. The expansion of the foam is the ratio of the volume of the foam to the volume of foam-liquid solution contained in the foam.

6.3.3 Foam. Foam as referred to in this specification is understood to denote mechanical foam produced for fire fighting purposes by the mechanical agitation of air with foam-liquid solution. It is an aggregation of small air-filled bubbles of lower specific gravity than oil or water and readily covers and clings to vertical or horizontal surfaces. Due to its high water retention ability, it cools hot surfaces and retains its properties for long periods of time. A fire fighting foam of good quality forms an air-excluding continuous blanket to seal off volatile combustible vapors from access to air or oxygen and thereby extinguishes the fire. This foam cover is long lasting and resists disruption due to wind and draft or heat and flame attack.

6.3.4 Foam-liquid solution. When AFFF concentrate conforming to MIL-PRF-24385 is mixed with water for use as a fire extinguishing agent, the solution thus formed is termed a foam-liquid solution for the purpose of this specification.

6.4 First article testing. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples from bidders offering a product which has been previously procured or tested by the Government. Bidders offering such products, who wish to rely on such procurement or test, must furnish evidence with the bid that prior Government approval is appropriate for the pending procurement.

6.5 Subject term (key word) listing.

AFFF

Aqueous Film Forming Foam

Aspirating

B-1

Fire Fighting

Nozzle

Overhead Sprinkler System

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH
Air Force – 11

Review activities:
Navy – MC
DLA – CC

Preparing activity:
Navy − SH
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